| Problem Chosen | 2026 | Team Control Number |
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| C | MCM/ICM | 2629912 |
| | Summary Sheet | |

# Reconstructing Hidden Popularity: A Hierarchical Bayesian Analysis of Judge–Audience Discrepancies in Competitive Elimination Systems


## Summary

In Dancing with the Stars (DWTS), elimination outcomes arise from the interaction of judges' evaluations and audience votes, despite the latter being unobserved. Against the backdrop of repeated controversy-driven changes between ranking-based and percentage-based elimination systems, we address five key questions:

Latent Fan-Vote Inference (Problem 1): We formulate fan support as a latent popularity process within a Bayesian framework, mapping relative popularity to weekly vote shares through a softmax structure. To capture season dynamics, we introduce a time-varying dispersion parameter that reflects shifts in voting concentration over time. Competition rules are incorporated through a rule-consistent likelihood linking latent votes to observed eliminations, and posterior inference via Markov Chain Monte Carlo (MCMC)provides uncertainty-aware estimates of fan voting behavior.

Mechanism Comparison via Counterfactual Simulation (Problem 2): Drawing fan votes from the posterior distribution, we replay each season under alternative aggregation rules using counterfactual simulations to isolate rule-driven effects. Beyond marginal responsiveness, we provide regression-based evidence disentangling the relative influence of fan votes and judges' scores. Using historically controversial contestants as fairness stress tests, we compare existing scoring systems and synthesize the results into principled recommendations.

Drivers of Judges vs. Fans (Problem 3): We construct parallel Mixed-Effects Regression models for judges' scores and inferred fan vote shares, accounting for season- and partner-level heterogeneity. We further validate nonlinear patterns using Random Forest models with SHAP-based interpretation. The results support a systematic influence of celebrity background characteristics on fan support and confirm a structural distinction between technical evaluation and popularity dynamics.

Voting System Design and Recommendation (Problem 4): We propose a new adaptive scoring system that integrates judge evaluations and fan votes by allowing the influence of fan voting to vary with its informational content. To assess whether the system achieves a better balance between fairness and entertainment, we introduce four evaluation metrics capturing technical fairness, perceived fairness, competitiveness, and late-stage uncertainty. The proposed system avoids extreme behaviors while achieving stronger fairness and comparable excitement.

Our findings highlight the complexity of competition outcomes arising from the interaction between technical evaluation and audience behavior, combining principled statistical inference with mechanism-level analysis. The proposed framework is also broadly applicable to other partially observed voting and ranking systems.

Keywords: DWTS, Bayesian Model, softmax, MCMC, voting mechanism, Mixed-Effects Regression, Random Forest, fairness–engagement trade-off, counterfactual simulation.

# 1 Introduction

## 1.1 Problem Background

In Dancing with the Stars (DWTS), celebrities paired with professional dancers receive combined scores from judges and audience votes. Judge scores reflect dance technique, while audience votes reflect viewer preferences, potentially influenced by fame and personal appeal. DWTS has employed various scoring systems, including ranking-based and percentage-based approaches. Due to controversial outcomes, the show has revised its rules multiple times. Since total viewer vote counts remain undisclosed, the extent to which the voting mechanism amplifies or mitigates audience influence remains unknown. Therefore, it is necessary to develop a quantitative framework to infer viewer voting behavior and assess the impact of different voting rules.

## 1.2 Clarifications and Restatement

This paper constructs a mathematical model to estimate the potential fan vote count for Dancing with the Stars and assesses the impact of different voting combination mechanisms on competition outcomes. We aim to evaluate the fairness, robustness, and sensitivity of existing rules while exploring potential improvement strategies. Our work consists of the following steps:

1. Estimate weekly fan votes and quantify uncertainty using observed outcomes.

2. Compare rank-based and percentage-based aggregation methods across seasons and contestants; Analyze historically controversial cases to evaluate the impact of alternative rules.

3. Examine how contestant and professional partner characteristics affect judges' scores and fan support.

4. Propose a revised voting system supported by quantitative evidence.

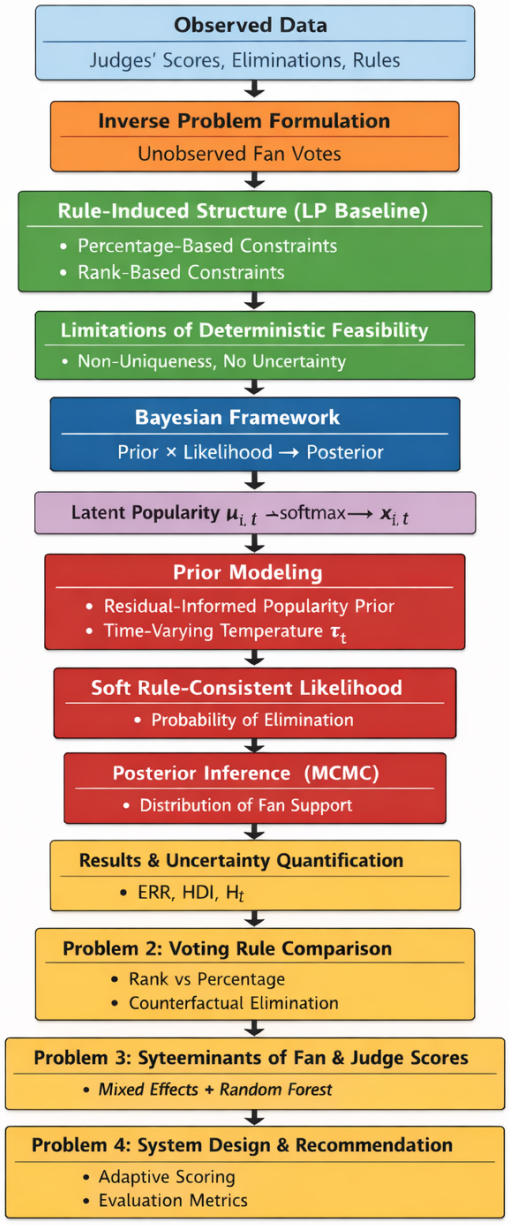
Observed Data
Judges' Scores, Eliminations, Rules
Inverse Problem Formulation
Unobserved Fan Votes
Rule-Induced Structure (LP Baseline)
• Percentage-Based Constraints
• Rank-Based Constraints
Limitations of Deterministic Feasibility
• Non-Uniqueness, No Uncertainty
Bayesian Framework
Prior × Likelihood → Posterior
Latent Popularity $\mu_{i,t}$ →softmax→ $x_{i,t}$
Prior Modeling
• Residual-Informed Popularity Prior
• Time-Varying Temperature $\tau_t$
Soft Rule-Consistent Likelihood
• Probability of Elimination
Posterior Inference (MCMC)
• Distribution of Fan Support
Results & Uncertainty Quantification
• ERR, HDI, $H_t$
Problem 2: Voting Rule Comparison
• Rank vs Percentage
• Counterfactual Elimination
Problem 3: Syteeminants of Fan & Judge Scores
• Mixed Effects + Random Forest
Problem 4: System Design & Recommendation
• Adaptive Scoring
• Evaluation Metrics

Figure 1:

### 1.3 Our Work

## 2 Preparation of Modeling

### 2.1 Model Assumption

- Judge rankings provide a reliable expert signal.
- Audience preference is a continuous quantity.
- Fan votes are noisy observations of latent preferences.
- Audience voting outcomes can be represented by normalized vote shares.
- Voting noise is predominantly random and evolves smoothly over time.
- Vote dispersion reflects informational content.
- Small vote perturbations approximate realistic uncertainty.
- Audience fairness perception depends on recent performance.
- Late-stage uncertainty contributes more to entertainment value.
- Contestants are evaluated independently within each week.

### 2.2 Notations

| Symbol | Description |
|---|---|
| $I_t$ | Set of contestants remaining in the competition at week $t$ |
| $J_{i,t}$ | Judges' score received by contestant $i$ in week $t$ |
| $x_{i,t}$ | Audience vote share of contestant $i$ in week $t$ |
| $y_{i,t}$ | Normalized judges' score share of contestant $i$ in week $t$ |
| $r^J_{i,t}$ | Rank of contestant $i$ based on judges' scores in week $t$ |
| $r^X_{i,t}$ | Rank of contestant $i$ based on audience votes in week $t$ |
| $z_{i,t}$ | Composite ranking score combining judges' and audience rankings |
| $e_t$ | Contestant eliminated in week $t$ |

Table 1: Notation Summary

# 3 Problem 1: Inferring Latent Fan Support from Elimination Outcomes

## 3.1 Latent Fan Voting as an Inverse Problem

To start the problem, we clarify that the observed elimination outcome $E_t$ depends on the combined effects of judge scores $J_{i,t}$, fan vote shares $x_{i,t}$, and competition rules. Since the mapping from fan votes to elimination outcomes is many-to-one, fan vote shares cannot be uniquely determined from observed data. Therefore, we attempt to describe a reasonable structural space for fan support rather than performing point estimation. Specifically, our goal is to identify the set of fan voting configurations that can produce elimination outcomes while describing the uncertainty and relative probabilities within that set.

## 3.2 Rule-Induced Structure of Fan Vote Configurations (LP Baseline)

(i) **Percentage-Based Constraints**

Under the percentage-based voting rule, elimination is determined by the combined contribution of fan vote shares and judges' scores, both normalized within each week. Specifically, for week $t$, judges' scores are normalized as

$$y_{i,t} = \frac{J_{i,t}}{\sum_{j \in I_t} J_{j,t}}, \quad i \in I_t. \tag{1}$$

and the total score used for ranking is defined as

$z_{i,t} := s_{i,t} = x_{i,t} + y_{i,t}. \quad z_{e_t,t} \leq z_{i,t}, \quad \forall i \in I_t.$

The eliminated contestant $E_t$ attains the lowest total score among all remaining contestants, yielding the ordering constraint. Together with the simplex constraint on fan vote shares $\sum_{i \in I_t} x_{i,t} = 1$ and $x_{i,t} \geq 0$, these inequalities characterize the set of fan vote configurations.

(ii) **Rank-Based Constraints** We considered ordinal information for rank-based voting rules. Judges' ranks $r^J_{i,t}$ are computed directly from observed scores $J_{i,t}$, thus $r^J_{i,t}$ is treated as fixed for each week $t$. Fan-vote ranks are induced by the unknown vote shares $x_{i,t}$. To encode the implied ordinal structure, we introduce pairwise order indicators:

$$b_{ij,t} = I\{x_{i,t} \geq x_{j,t}\} \tag{2}$$

and enforce consistency through standard Big-M linear constraints

$$x_{i,t} - x_{j,t} \geq \varepsilon - M(1 - b_{ij,t}), \tag{3}$$

$$x_{j,t} - x_{i,t} \geq \varepsilon - M b_{ij,t}. \tag{4}$$

where $M$ is sufficiently large and $\varepsilon > 0$ ensures strict ordering. Given these indicators, the fan-vote rank and the combined rank is

$$r^X_{i,t} = 1 + \sum_{\substack{j \in I_t \\ j \neq i}} (1 - b_{ij,t}), \quad z_{i,t} = r^J_{i,t} + r^X_{i,t}, \quad z_{e_t,t} \geq z_{i,t}, \quad \forall i \in I_t. \tag{5}$$

The eliminated contestant is the one with the largest combined rank, implying the ordering condition. Together with the simplex constraint on fan vote shares, these inequalities characterize the set of fan vote configurations. Collectively, in this baseline model, linear programming serves as a conceptual tool to reveal the structural effects of competitive rules. However, the linear programming benchmark model treats all feasible configurations as equally probable and does not provide uncertainty quantification, prompting us to undertake a probabilistic extension.

## 3.3 From Feasibility to Probability: A Bayesian Framework

To overcome these limitations, we adopt a Bayesian framework, reformulating the inference problem as a probabilistic one. Let D denote the observed data (including judges' scores, elimination outcomes, and competition rules), and $\Theta$ denote the set of latent variables and model parameters. Bayesian inference aims to solve for the posterior distribution

$$p(\Theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \Theta)\, p(\Theta), \tag{6}$$

where $p(\Theta)$ encodes prior assumptions on latent popularity and its evolution, and $p(D \mid \Theta)$ represents the likelihood that the observed eliminations arise from these latent quantities under the competition rules. In this framework, the deterministic ordering structures derived in LP are embedded into the likelihood in a soft, probabilistic manner. Bayesian model assigns probability mass over latent popularity trends that are more or less consistent with the observed elimination outcomes. The following sections specify the components of this framework in detail.

## 3.4 Latent Popularity Scores and Softmax Mapping to Fan Votes

Directly modeling fan vote shares $x_{i,t}$ is challenging due to the simplex constraints and the relative nature of voting outcomes. Instead, we introduce latent popularity scores $u_{i,t} \in R$,

which represent the underlying strength of fan support for contestant i in week t and have no constraint. They serve as the fundamental random variables in our Bayesian model, but are not directly observable. To map latent popularity scores to valid vote shares, we apply the softmax transformation:

$$x_{i,t} = \frac{\exp(u_{i,t})}{\sum_{j \in I_t} \exp(u_{j,t})}, \quad i \in I_t. \tag{7}$$

This mapping ensures that $x_{i,t} \geq 0$ and $\sum_{i \in I_t} x_{i,t} = 1$ automatically and a higher latent popularity corresponds to a larger expected vote share. The softmax mapping is invariant to adding the same constant to all $u_{i,t}$ within a week. Hence u is not identifiable without a reference constraint. To remove this redundancy and improve numerical stability, we impose a within-week centering constraint for each week t $\sum_{i \in I_t} u_{i,t} = 0$. All results regarding vote shares $x_{i,t}$ are invariant to the chosen gauge.

## 3.5 Data-Informed Prior for Latent Popularity

### 3.5.1 Residual-Based Popularity Signal

To extract additional information about latent fan support, we construct a residual-based signal using observed elimination outcomes. Specifically, let $S^i$ denote the number of weeks survived of contestant $i$, and let $\hat{S}^i(J)$ denote the survival length predicted using judges' scores alone. We define the standardized residual

$$r_i = \frac{S_i - \hat{S}_i(J)}{\mathrm{sd}(S - \hat{S})}. \tag{8}$$

A positive residual indicates that contestant i outperformed expectations based solely on judges' evaluations, suggesting stronger fan support. Figure 3.5A plots normalized survival outcomes against average judges' scores, and contestants with large positive residuals are highlighted. These cases represent contestants whose longevity cannot be fully explained by judges' scores, suggesting elevated latent popularity. Figure 3.5B displays the distribution of standardized residuals obtained from a linear regression of survival length on judges' scores. A right tail corresponds to contestants who survived substantially longer than predicted by judges' score alone, which provide evidence of audience-driven support.

### 3.5.2 Residual-Informed Prior Specification

We incorporate this residual signal as a weakly data-informed prior on latent popularity. For each contestant i and week t, we specify

$$u_{i,t} \sim \mathcal{N}(cr_i, \sigma_u^2). \tag{9}$$

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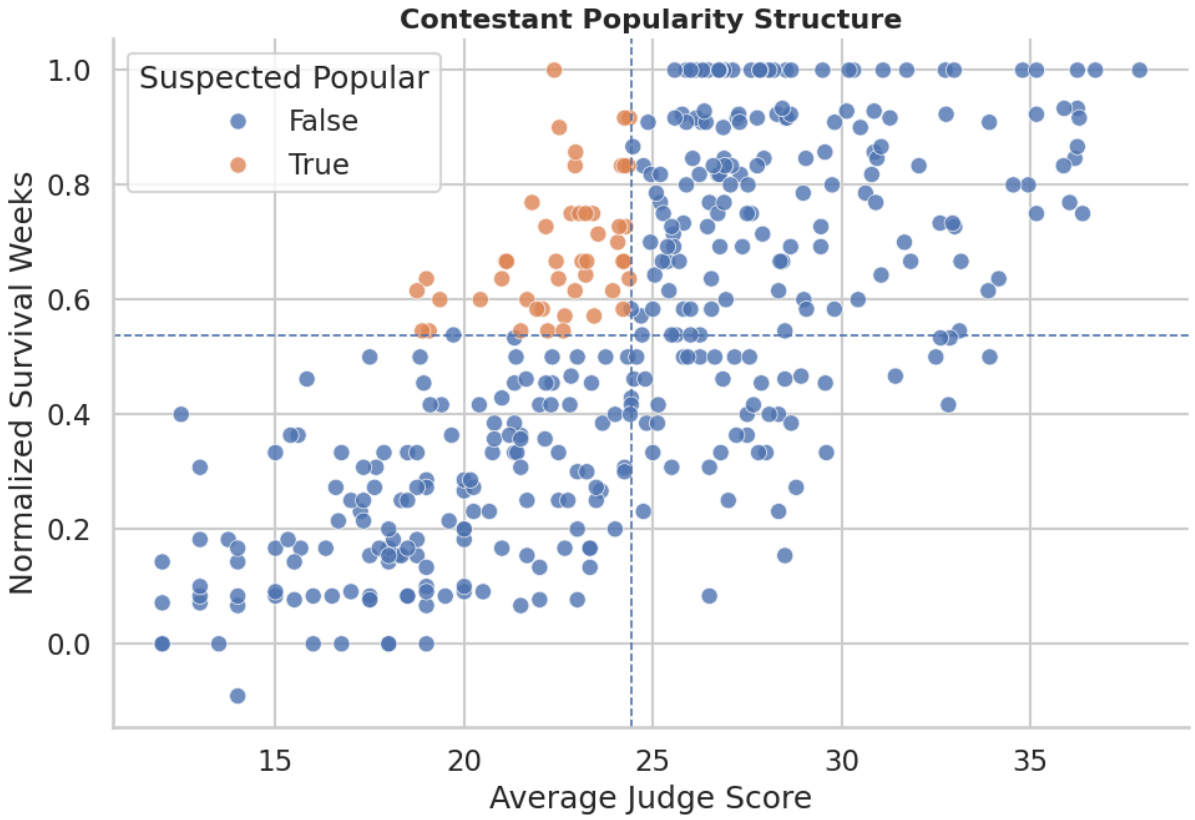


Figure 2: Contestant Popularity Structure

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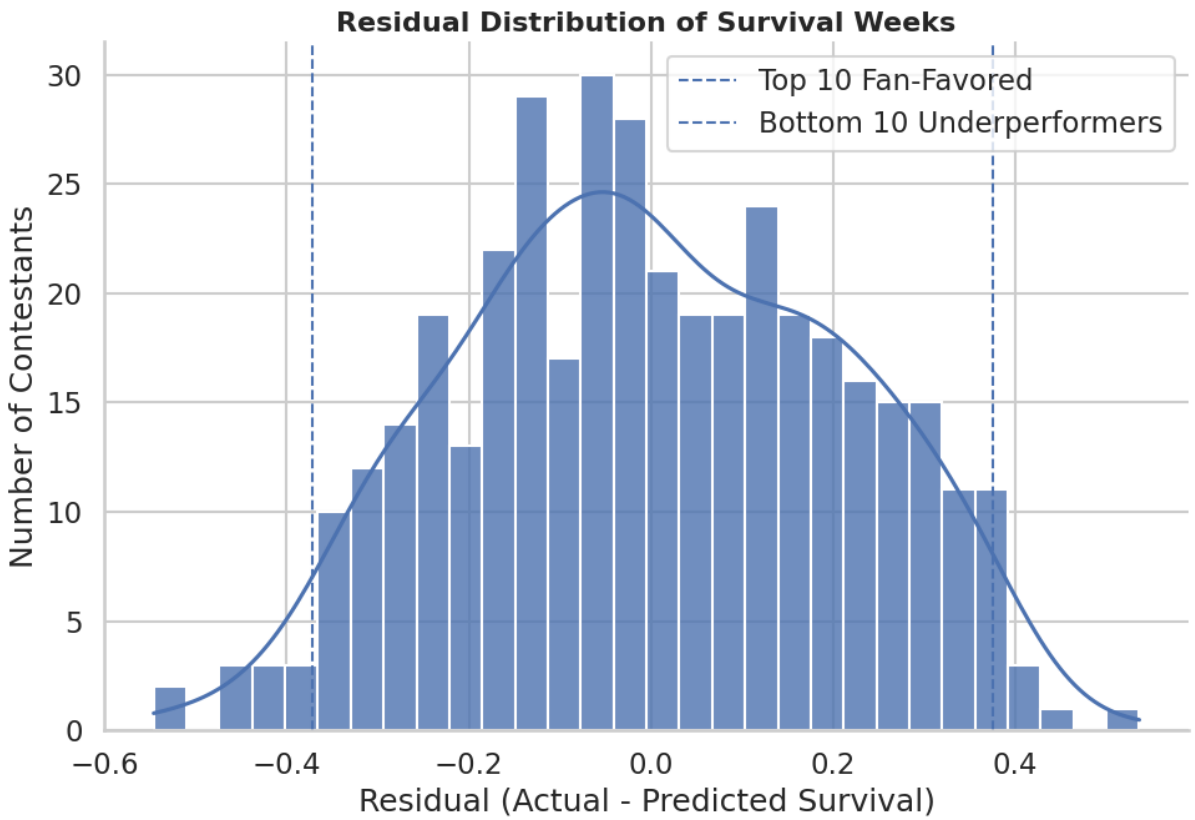


Figure 3: Residual Distribution

Figure 4: Fan Vote Structure and Residual Distribution

where $c \geq 0$ controls the strength of the residual effect. Setting $c = 0$ yields a neutral baseline prior, while $c > 0$ allows contestants who outperform judge-based expectations to have a higher prior mean popularity. To mitigate potential information leakage, we assess robustness by varying the strength of this prior in sensitivity analyses.

## 3.6 Time-Varying Voting Stochasticity via Softmax Temperature

### 3.6.1 Temperature as Voting Stochasticity

Latent popularity reflects underlying fan preference, but observed voting behavior may exhibit additional randomness. To decouple preference strength from voting stochasticity, we introduce a temperature parameter $\tau_t > 0$ into the softmax mapping:

$$x_{i,t} = \frac{\exp(u_{i,t}/\tau_t)}{\sum_{j \in I_t} \exp(u_{j,t}/\tau_t)}. \tag{10}$$

Where larger $\tau_t$ flattens vote shares, indicating more diffuse or random voting, while a smaller $\tau_t$ amplifies popularity differences and yields more concentrated outcomes.

### 3.6.2 Logistic Time Dynamics of Temperature

Voting behavior may evolve over the course of a season. Early stages often exhibit exploratory or scattered voting patterns, while later stages see voting trends stabilize as audience preferences solidify. To capture this effect, we model voting temperature as a logical function of time:

$$\tau_t = \tau_{\min} + \frac{\tau_{\max} - \tau_{\min}}{1 + \exp\big(\kappa(t - t_{\text{pivot}})\big)}, \tag{11}$$

$$\Delta_\tau = \tau_{\max} - \tau_{\min} > 0. \tag{12}$$

The temperature dynamics allow voting stochasticity to vary smoothly over time without introducing abrupt regime changes. Here, $\tau_{\max}$ and $\tau_{\min}$ represent the upper and lower bounds of voting stochasticity, $t_{pivot}$ denotes the transition point in the season, and $\kappa > 0$ controls the rate of change, with the effective transition width inversely proportional to $\kappa$, allowing $\kappa$ to be interpreted as a smoothness parameter. To ensure positivity and improve sampling efficiency, we adopt the reparameterization. Priors are assigned to $\tau_{\min}$, $\Delta\tau$, and $\kappa$ to reflect scale constraints while remaining weakly informative. Parameter choices and prior specifications were guided by data scale and sampling stability considerations; Implementation details and sensitivity analyses are provided in Appendix.

## 3.7 Likelihood: Soft Rule-Consistent Elimination Mechanism

In our Bayesian framework, competition rules are incorporated through a soft likelihood rather than hard feasibility constraints. Given latent popularity $u_t$, temperature $\tau_t$, and the implied fan vote shares $x_t$, the likelihood assigns higher probability to outcomes that are more consistent with the competition rules, while still allowing limited stochasticity. This formulation preserves the spirit of the deterministic LP constraints while enabling probabilistic inference and uncertainty quantification. We define the likelihood of observing the eliminated contestant $E_t$ as

$$\Pr(e_t = i \mid z_t) = \frac{\exp(\beta z_{i,t})}{\sum_{j \in I_t} \exp(\beta z_{j,t})}. \tag{13}$$

where $\beta > 0$ controls the sharpness of rule enforcement. Larger values of $\beta$ place increasing probability mass on the lowest-ranked contestants, approximating deterministic elimination, while smaller values allow greater randomness. Competition rule differences in likelihood formulation enter only through the latent ordering score $z_{i,t}$ as defined in LP. Thus, the LP baseline defines the structural logic of elimination, while the soft likelihood embeds this logic into a fully probabilistic framework.

## 3.8 Posterior Inference and Sampling

Given the prior structures and the soft rule-consistent likelihood defined above, the Bayesian model induces a posterior distribution over latent popularity trends and auxiliary parameters.

$$\Theta = \{u_{i,t} : i \in I_t,\ t = 1, \ldots, T\}, \tag{14}$$

$$p(\Theta \mid \mathcal{D}) \propto \prod_{t=1}^{T} p(e_t \mid z_t(\Theta))\ p(\Theta). \tag{15}$$

$\Theta$ denotes the collection of latent popularity scores across all contestants and weeks. Conditional on $\Theta$ and the temperature parameters $\tau_t$, fan vote shares $x_{i,t}$ are deterministically obtained via the softmax mapping. The posterior distribution is given by $P(\Theta \mid \mathcal{D})$ where $P(\Theta)$ captures temporal structure and heterogeneity, and $P(e_t \mid z_t(\Theta))$ assigns higher probability to eliminations more consistent with the rule-induced ordering scores. As this posterior does not admit a closed-form expression, we perform posterior inference using Markov Chain Monte Carlo (MCMC) sampling. The resulting posterior draws provide a distribution over latent fan support patterns. These samples enable quantification of uncertainty and provide a basis for further analyses.

## 3.9 Results and Uncertainty Quantification

### 3.9.1 Consistency with Observed Eliminations

Posterior samples $\{\Theta^{(s)}\}_{s=1}^{S}$ rom the MCMC imply weekly fan vote shares $x_t^{(s)}$ through the softmax mapping. We evaluate whether these inferred vote shares are consistent with observed eliminations using an outcome-based posterior predictive check. For each week $t$ and draw $s$, the predicted eliminated contestant $\hat{e}_t^{(s)}$ is determined by the elimination rule. Define the Elimination Reproduction Rate, and a larger ERR indicates stronger consistency between inferred fan support and observed outcomes:

$$\mathrm{ERR} = \frac{1}{T}\sum_{t=1}^{T}\left[\frac{1}{S}\sum_{s=1}^{S}\mathbf{1}\left\{\hat{e}_t^{(s)} = e_t^{\mathrm{obs}}\right\}\right]. \tag{16}$$

By conducting an outcome-based posterior predictive check and computing the Elimination Reproduction Rate (ERR), we find that the model successfully reproduces the observed eliminations in 73% of cases. It suggests strong but not perfect alignment between the latent support structure and actual outcomes. This reflects both the explanatory power of the model and the inherent uncertainty in elimination dynamics.

### 3.9.2 Certainty of Fan Vote Estimates

Uncertainty in fan vote percentages is quantified directly from posterior samples. For each contestant i in week t, we compute the 90Narrow HDIs indicate stable and inferred support, whereas wider HDIs reflect greater posterior uncertainty in the estimated vote percentages. To capture heterogeneity in certainty across contestants, we summarize each week by $H_t$, defined as the difference between the maximum and minimum HDI widths among remaining contestants, thereby measuring dispersion in posterior uncertainty.

In Figure 2, the solid line shows the mean weekly uncertainty heterogeneity $H_t$ averaged across all seasons, while the shaded region indicates variability across seasons. In the early stage of the competition, $H_t$ remains low, indicating that posterior uncertainty is relatively uniform across contestants when voting behavior is highly diffuse and little information is available. As the season progresses, uncertainty heterogeneity increases sharply and reaches a peak in the mid-season. This pattern suggests that while support for some contestants gradually solidified, significant posterior uncertainty persisted for others. It reflects asymmetric information accumulation, where fan preferences consolidate unevenly across contestants. Toward the final stage, $H_t$ declines but remains elevated relative to the beginning of the season. Despite overall contraction as contestant numbers decrease, ambiguous fan support persists for certain contestants, making their relative rankings more sensitive to minor shifts in inferred vote shares. The shaded bands further reveal pronounced differences across seasons, underscoring that while the temporal pattern remains consistent, the degree of uncertainty heterogeneity depends on

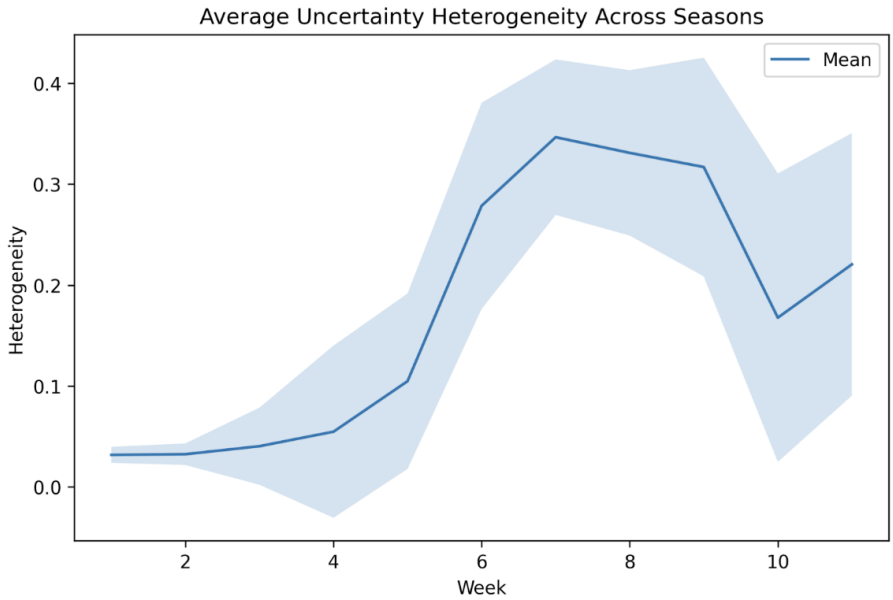


Figure 5:

specific seasonal characteristics.

# 4 Problem 2: Application of Model Results: Voting Mechanism Analysis

## 4.1 Implementation Across Seasons Using Estimated Fan Votes

Applying both aggregation methods to our estimated weekly vote distributions from Problem 1, we find that the majority of elimination outcomes, 176 cases (approximately the majority), are consistent across systems. However, 75 cases exhibit outcome reversals. Among these reversals, 47 contestants retained under the rank-based system are eliminated when percentage aggregation is applied, and they exhibit relatively low fan vote shares (average 5.9%) despite moderate judge score percentages. In contrast, 28 contestants who would have been eliminated under the rank-based system are retained under the percent-based system, and they receive substantially higher fan vote shares (average 11.0%), even though their judge score percentages are slightly lower. These discrepancies suggest that the percent-based method systematically alters outcomes relative to rank aggregation, particularly in cases where fan vote distributions are uneven.

## 4.2 Do Voting Systems Favor Fan Votes?

A voting system is said to favor fan votes if changes in fan support are more strongly reflected in final outcomes, holding judges' evaluations fixed. Under this definition, favoring fan votes

refers to the responsiveness at the margin or contributes more strongly to outcome variation.

### 4.2.1 A Marginal Utility Perspective

To provide a brief structural comparison, we examine how marginal changes in fan support affect outcomes under different aggregation mechanisms. Under the rank-based system, outcomes depend only on contestants' ordinal positions. In a week with $n_t$ remaining contestants, fan votes can alter results only at at most $n_{t-1}$ rank-switching thresholds. Consequently, the marginal utility of fan votes under rank aggregation is zero almost everywhere with respect to vote counts. Under the percent-based system, outcomes depend on vote magnitudes. Let $f_i$ denote fan votes for contestant $i$, with vote share

$$p_i^{\text{fan}} = \frac{f_i}{\sum_j f_j}. \tag{17}$$

Because the combined score depends continuously on $p_i^{fan}$, incremental increases in $f_i$ always increase the contestant's score, holding judges' scores fixed. In summary, rank aggregation compresses fan influence into discrete events, whereas percent aggregation preserves continuous responsiveness to fan support. This distinction characterizes the potential influence of fan votes at the mechanism level.

### 4.2.2 Regression-Based Evidence of Fan and Judge Influence

To complement the structural analysis above, we use regression models to summarize how judges' evaluations and fan support are reflected in final outcomes. For each voting system, we estimate an ordinary least squares (OLS) regression of the form:

$$y_i = \alpha + \beta_J J_i + \beta_F F_i + \varepsilon_i. \tag{18}$$

where $y_i$ denotes the final outcome for contestant $i$, $J_i$ the judges' input, and $F_i$ the fan input. Under the rank-based system, these variables correspond to final rank, judges' rank, and fan vote rank. Under the percent-based system, they correspond to the final combined percentage, the judges' score percentage, and the fan vote percentage. All variables are standardized using z-scores before estimation, allowing coefficient magnitudes to be interpreted as relative contributions of judges and fans within each system.

The regression results reveal distinctly different patterns across systems and highlight how aggregation rules fundamentally determine the mapping from input to output:

In the ranking-based system, the estimated coefficients for judge rankings and fan vote rankings are similar in magnitude. Both input variables exhibit strong explanatory power, and the model achieves high goodness-of-fit. From an interpretive perspective, neither party systematically dominates the final ranking. The ordinal transformation merely compresses judge

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OLS Regression Results

| | | | |
|---|---|---|---|
| Dep. Variable: | y | R-squared: | 0.940 |
| Model: | OLS | Adj. R-squared: | 0.940 |
| Method: | Least Squares | F-statistic: | 2.142e+04 |
| Date: | Mon, 02 Feb 2026 | Prob (F-statistic): | 0.00 |
| Time: | 01:28:36 | Log-Likelihood: | -33.879 |
| No. Observations: | 2738 | AIC: | 73.76 |
| Df Residuals: | 2735 | BIC: | 91.50 |
| Df Model: | 2 | | |
| Covariance Type: | nonrobust | | |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| const | -4.425e-16 | 0.005 | -9.45e-14 | 1.000 | -0.009 | 0.009 |
| x1 | 0.5715 | 0.005 | 113.342 | 0.000 | 0.562 | 0.581 |
| x2 | 0.5999 | 0.005 | 118.980 | 0.000 | 0.590 | 0.610 |

| | | | |
|---|---|---|---|
| Omnibus: | 19.412 | Durbin-Watson: | 1.977 |
| Prob(Omnibus): | 0.000 | Jarque-Bera (JB): | 21.072 |
| Skew: | -0.163 | Prob(JB): | 2.66e-05 |
| Kurtosis: | 3.279 | Cond. No. | 1.47 |

Figure 6: Table: OLS regression results under rank-based aggregation

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OLS Regression Results

| | | | |
|---|---|---|---|
| Dep. Variable: | y | R-squared: | 0.897 |
| Model: | OLS | Adj. R-squared: | 0.897 |
| Method: | Least Squares | F-statistic: | 1.195e+04 |
| Date: | Mon, 02 Feb 2026 | Prob (F-statistic): | 0.00 |
| Time: | 01:28:43 | Log-Likelihood: | -769.12 |
| No. Observations: | 2738 | AIC: | 1544. |
| Df Residuals: | 2735 | BIC: | 1562. |
| Df Model: | 2 | | |
| Covariance Type: | nonrobust | | |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| const | -6.551e-16 | 0.006 | -1.07e-13 | 1.000 | -0.012 | 0.012 |
| x1 | 0.3436 | 0.007 | 52.102 | 0.000 | 0.331 | 0.357 |
| x2 | 0.7648 | 0.007 | 115.962 | 0.000 | 0.752 | 0.778 |

| | | | |
|---|---|---|---|
| Omnibus: | 265.740 | Durbin-Watson: | 2.153 |
| Prob(Omnibus): | 0.000 | Jarque-Bera (JB): | 1841.736 |
| Skew: | -0.137 | Prob(JB): | 0.00 |
| Kurtosis: | 7.009 | Cond. No. | 1.47 |

Figure 7: Table: OLS regression results under percent-based aggregation

Figure 8: Fan Vote Structure and Residual Distribution

and fan information into relative positions, limiting the influence of vote counts.

In contrast, within the percentage-based system, the coefficient for fan vote percentage is significantly larger than that for judge score percentage. The model retains high Differences in fan support directly translated into differences in composite scores, amplifying the influence of fan votes on the final outcome. This amplification effect manifests as a larger standardization coefficient for fan vote percentages.

It is important to clarify that these regression analyses are not intended to establish causality but serve as diagnostic tools to illustrate how different aggregation mechanisms utilize information from judges and fans. These differences directly inform that a percentage-based voting system can be said to favor fan votes in practice.

## 4.3 Controversial Contestants as Stress Tests- When Do Mechanisms Matter More?

Although alternative aggregation rules yield identical elimination outcomes in most weeks, a small but consequential subset of contestants exhibits persistent disagreement between judges' evaluations and fan support. These cases serve as natural stress tests for voting mechanisms: if rule choice matters anywhere, it should matter most when technical assessment and audience preference diverge.

### 4.3.1 Identification of Controversial Cases

Using the estimated posterior fan vote distributions from section 3, we identify controversial contestants based on two criteria:

1. Judge–Fan Divergence: inferred fan vote share lies in the upper tail of the posterior distribution while normalized judges' scores remain persistently low.

2. Outcome Sensitivity: elimination status differs across rank-based and percentage-based aggregation

### 4.3.2 Counterfactual Outcomes

To evaluate how aggregation rules affect controversial contestants, we conduct a counterfactual analysis. Holding inferred fan vote shares and observed judges' scores fixed, we re-compute weekly eliminations under: rank-based aggregation, percentage-based aggregation, and rank aggregation combined with the "bottom-two + judges' save" rule.

Several systematic patterns emerge:

- Fan-amplification under percentage aggregation: Contestants such as Jerry Rice and Bobby Bones, characterized by weak technical scores but strong fan bases, perform substantially better under percentage-based rules. It aligns with the earlier statement that marginal increases in fan votes translate continuously into higher combined scores.

| Season | Contestant | Low Judge Scores? | Strong Fan Support? | Rank Outcome | Percent Outcome | Judges Save Effect |
|---|---|---|---|---|---|---|
| 2 | Jerry Rice | Yes (5 weeks) | High | Advance | Win | Safe |
| 4 | Billy Ray Cyrus | Yes (6 weeks) | Moderate | Advance | Advance | Likely eliminated |
| 11 | Bristol Palin | Yes (12 weeks) | High | Advance | Advance(stronger) | Likely Eliminated |
| 27 | Bobby Bones | Consistently low | High | Advance (3rd place) | Win | Likely Eliminated |

Figure 9: Counterfactual outcomes under alternative rules

- Rank-based buffering effects: Rank aggregation compresses vote magnitudes into ordinal positions. As a result, contestants with middling relative ranks but low absolute fan shares—such as Billy Ray Cyrus—experience relatively small changes across rules.
- The introduction of a "bottom two elimination system" can significantly reduce instances where contestants advance solely through fan preferences—especially when such contestants consistently receive low scores from the judges. Under this mechanism, even the four controversial contestants, despite their strong fan bases, remain highly likely to face elimination. These findings suggest that mechanism selection selectively amplifies or suppresses information from different sources based on how fan votes enter the aggregation process.

#### 4.3.3 Judges' Save and the Bottom-Two Rule

The judge rescue mechanism introduces an intervention point into the elimination process, triggering only when results fall within a narrow range of high uncertainty—namely, the bottom two contestants. From a modeling perspective, this rule functions as a conditional override: audience votes determine which contestants enter the risk pool, while judges retain decisive control over the final elimination outcome. In controversial cases, this mechanism effectively truncates the upper tail of fan influence, preventing extreme audience mobilization from overturning persistently low professional scores. The case study above indicates that the judge rescue mechanism significantly impacts specific contestants—namely those posterior fan support variance is high, but expected judges' score remains low. This precisely matches the defining characteristics of historically controversial cases.

### 4.4 Recommendations for Future Seasons

Based on a counterfactual analysis of aggregation rules, regression analysis, marginal responsiveness, and controversial cases, we recommend adopting ranking-based aggregation as the default voting system, with judges choosing from the bottom two couples in the later stages of the competition. Our findings indicate that ranking-based aggregation provides a structurally stable baseline solution: by relying on ordinal information, this mechanism both mitigates the amplification effect of extreme fan mobilization and maintains a balance between judge and fan contributions. In contrast, percentage aggregation excessively amplifies concentrated fan support, making outcomes more susceptible to disruption when judge and fan evaluations diverge. Contested contestant cases reveal that instability does not come from fan participation itself, but from the absence of targeted corrective mechanisms. Introducing a judge's save partially resolves this issue under specific conditions—triggered only when elimination outcomes fall within a high-risk narrow margin—avoiding persistent interference with audience influence. Therefore, the hybrid design—using ranking aggregation for regular weeks and judge save during controversial elimination rounds—achieves a principled balance between fairness, robustness, and entertainment value. This mechanism maintains audience engagement while preventing systemic biases caused by extreme popularity, aligning closely with the long-term credibility goals of the competition format.

## 5 Problem 3: Impact of Celebrity and Partner Characteristics on Judges' Scores and Fan Votes

In this section, we want to examine whether observable celebrity characteristics influence judges' scores and fan votes in the same way. While judges are expected to evaluate technical dance performance, fan votes may reflect popularity, personal appeal, or background-related preferences. Distinguishing between these two evaluation mechanisms is essential for understanding the sources of disagreement observed in several seasons.

### 5.1 Mixed-Effects Modeling Framework

A key challenge arises from the structure of the data: each celebrity appears multiple times across weeks, professional dancers are paired with multiple celebrities across seasons, and scoring environments vary by season. As a result, observations are not independent, and simple linear regression would risk confounding celebrity characteristics with partner-specific or season-specific effects. To address this hierarchical structure, we adopt a mixed-effects modeling framework that allows us to isolate the average effect of celebrity characteristics while controlling for persistent differences attributable to professional partners and seasons.

### 5.1.1 Model Specification

We estimate two parallel linear mixed-effects models: one for judges' evaluations and one for fan voting outcomes. Both models share the same explanatory structure, ensuring that differences in results reflect differences in evaluation mechanisms rather than differences in model specification. Let $i$ index celebrities and $t$ index weeks.

$$\text{JudgeScore}_{i,t} = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + u_{\text{partner}(i)} + v_{\text{season}(i)} + \varepsilon_{i,t}, \tag{19}$$

$$\text{FanShare}_{i,t} = \alpha' + \boldsymbol{\beta}'^\top \mathbf{X}_i + u'_{\text{partner}(i)} + v'_{\text{season}(i)} + \varepsilon'_{i,t}. \tag{20}$$

where

- $\mathbf{X}_i$ includes observable celebrity characteristics, specifically industry category and age,
- $u_{\text{partner}(i)}$ is a random intercept capturing persistent effects associated with the professional dance partner,
- $v_{\text{season}(i)}$ is a random intercept accounting for season-level differences in scoring environments, and
- $\varepsilon_{i,t}$is an idiosyncratic error term.

The fixed effects $\beta$ and $\beta'$ capture the average association between celebrity characteristics and outcomes, while the random effects absorb unobserved heterogeneity due to partner skill and season-specific factors.

### 5.1.2 Results and Interpretations

Results are reported relative to the baseline category, thus coefficients on other industry indicators represent average differences in judges' scores or fan vote shares relative to celebrity contestants from the acting profession, after controlling for partner and season effects. Our primary interest lies in the direction, relative size, and statistical significance of coefficients across the two models. By comparing $\boldsymbol{\beta}$and $\boldsymbol{\beta}'$, we assess whether the same celebrity characteristics are rewarded or penalized similarly by judges and fans.

(i) **Industry Effects on Judges' Scores** Average scores for some non-performing arts industries fell below baseline levels, suggesting that judges' assessments are not entirely immune to industry background influences. Notably, the overall dispersion of industry coefficients in the judge model is modest, suggesting that judge scoring exhibits relatively lower sensitivity to celebrity background compared to other controlled factors. This aligns with the expectation that judges evaluate technical performance rather than personal image.

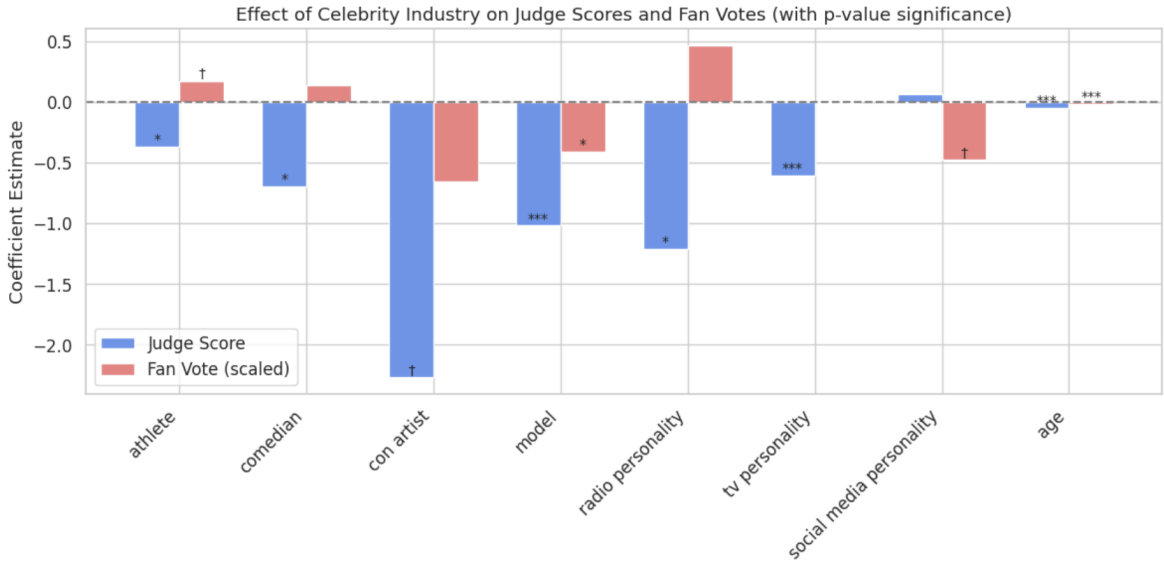


Figure 10: Estimated industry effects on judges' scores and fan vote shares (baseline: actor/actress)

(ii) **Industry Effects on Fan Vote Shares** In contrast, the fan voting model exhibits greater variability across industry categories. Some industries show statistically significant differences in voting shares relative to actors/actresses, with effects larger than those observed in the judge model. This suggests fan voting behavior is more strongly linked to celebrity background, consistent with the expectation that audience preferences reflect narrative appeal beyond popularity or technical skill. Industries associated with public-facing or identity-driven personal personas often exhibit greater deviations from baseline in fan support.

(iii) **Comparison**

Consistent Effects: some industries—such as models—are associated with lower outcomes in both judges' scores and fan votes, indicating shared negative perceptions.

Divergent Effects: For several industries—such as radio personality and television personality backgrounds—coefficients differ markedly across models, often negative or insignificant in judges' scores but positive and significant in fan vote shares.

Fan-Dominated Effects: Certain industries-military and motivational speakers— primarily affect fan voting outcomes while having little measurable association with judges' scores, suggesting that industry background influences audience engagement more than technical assessment. Taken together, these patterns demonstrate that the celebrity industry does not impact judges' scores and fan votes in the same way. The same background characteristics that generate additional fan support may have limited or even opposing associations with judges' evaluations.

## 5.2 Robustness Check Using Random Forests

### 5.2.1 Model Setup

To complement the mixed-effects analysis and relax linearity assumptions, we employ random forest regression models to examine the determinants of judges' scores and fan vote shares. Random forests are well-suited for capturing nonlinear relationships and interactions among variables, but do not yield directly interpretable coefficients. To address this limitation, we use SHAP (SHapley Additive exPlanations) values to interpret feature contributions. Two separate random forest models are trained with identical feature sets to predict the average judges' score and the scaled fan vote share. SHAP values decompose the prediction of each observation into additive contributions from individual features. We focus on mean absolute SHAP values, which measure the average magnitude of each feature's contribution across all samples.

### 5.2.2 Random Forest Results for Judges' Scores

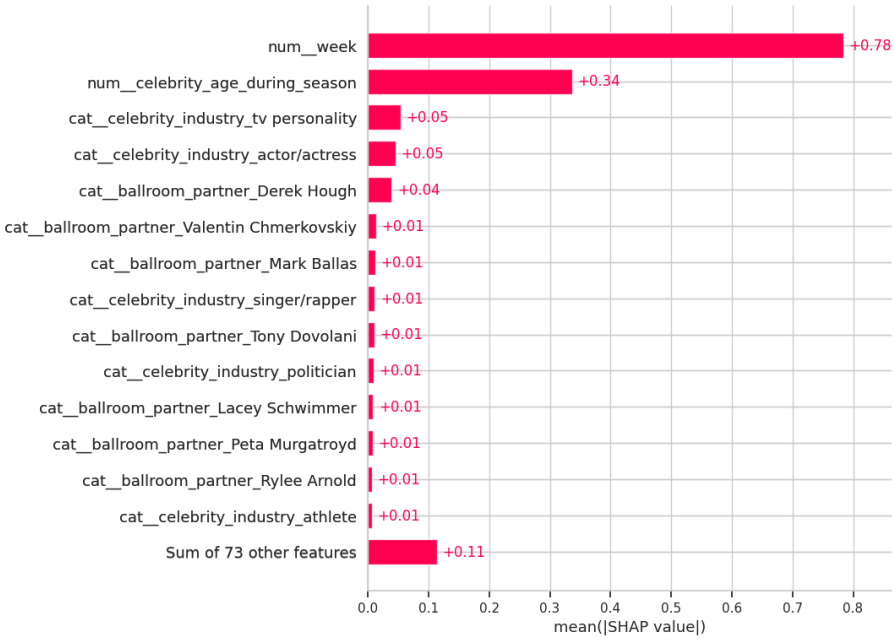


Figure 11:

From Figure 6 we can see that the most influential feature is the competition week. This reflects either cumulative improvement over time or, more plausibly, a selection effect whereby higher-scoring contestants remain in later weeks. The second most influential feature by a large margin is celebrity age during the season, suggesting that age systematically affects judges' evaluations, potentially through performance ability, presentation, or narrative emphasis. Industry and professional partner variables exhibit smaller but non-negligible contributions. Among industry categories, actor/actress and television personality display the largest SHAP values, while other industries, such as singer/rapper, politician, and athlete, have more

limited individual effects. Several professional partners, including Derek Hough and Valentin Chmerkovskiy, also appear among the more influential categorical features, indicating persistent partner-related effects. Finally, the aggregate contribution of numerous low-impact features ("Sum of 73 other features") is nontrivial, highlighting the presence of many small influences that collectively affect judges' scores.

### 5.2.3 Random Forest Results for Fan Vote Shares

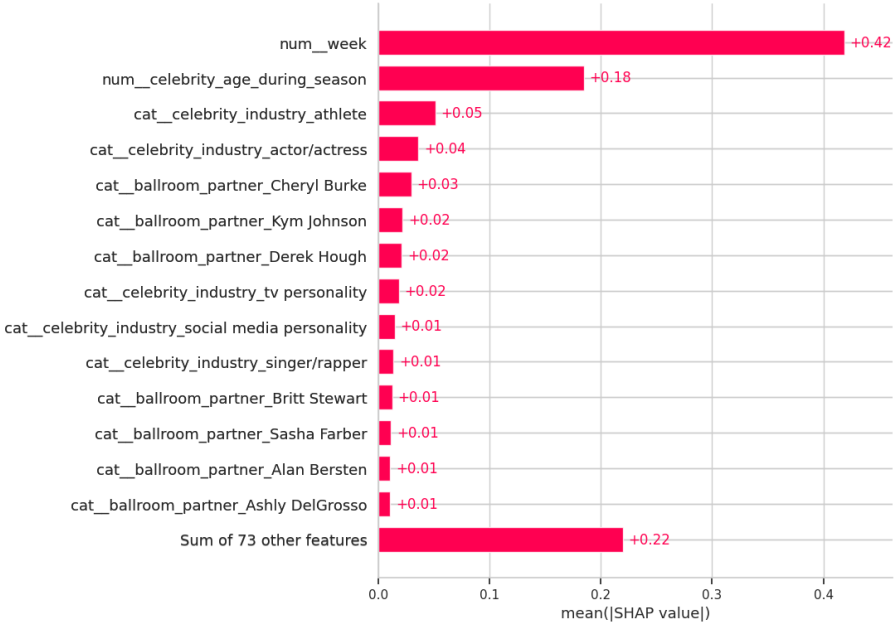


Figure 12:

Figure 7 shows that, as in the judges' model, competition week is the most influential predictor, followed by celebrity age. However, compared with judges' scores, celebrity background characteristics play a more prominent role. In particular, the athlete and actor/actress industry indicators exhibit higher SHAP values, suggesting that fan voting behavior is more sensitive to celebrity identity and background. Several professional partners, including Cheryl Burke, Kym Johnson, and Derek Hough, also contribute modestly to fan vote predictions, consistent with the idea that well-known partnerships may enhance audience appeal. Other industry categories, such as television personality, social media personality, and singer/rapper, display smaller individual SHAP values, indicating weaker but still measurable effects on fan support. As in the judges' model, a large number of low-impact features collectively contribute a meaningful share of predictive power.

## 5.3 Integrated Interpretation

Across both models, temporal progression is the dominant driver of outcomes. However, celebrity characteristics—especially industry and age—play a more prominent role in predict-

ing fan votes than judges' scores. This asymmetry persists in a nonlinear, nonparametric setting, confirming that the differences observed in the mixed-effects analysis are not artifacts of linear model assumptions. Accordingly, the random forest and SHAP results provide robust support for the conclusion that judges and fans systematically weigh celebrity characteristics differently.

# 6 Problem 4: Developing a New Evaluation System

## 6.1 What Does "Better" Mean? — Evaluation Philosophy

Previous analysis shows that no scoring system in an entertainment-based competition can be evaluated solely by technical accuracy or audience influence. Systems that overly amplify audience preferences may undermine professional legitimacy, while systems that rely exclusively on expert judgment risk reducing audience engagement and suspense.

As a result, fairness and excitement are not opposing objectives but interacting dimensions that jointly shape perceived legitimacy. A scoring system that appears mechanically fair may still be viewed as unreasonable by audiences, while a system that maximizes entertainment value can erode competitive credibility.

Therefore, before proposing a new evaluation framework, we first clarify what "better" means in this context. We assess scoring systems along three complementary dimensions: technical fairness, perceived fairness, and entertainment value.

1. Technical fairness: A necessary foundation for competitive legitimacy. This refers to the extent to which a scoring system preserves the relative ordering implied by professional evaluations, ensuring that stronger contestants are not systematically ranked below weaker ones in the final outcomes, consistent with the principle of order preservation.

   Accordingly, we operationalize technical fairness by measuring the degree of order preservation through the Spearman rank correlation between judge rankings and system outcomes, which directly quantifies how consistently the relative ordering implied by professional evaluations is retained in the final results.

   $$F1_t = \rho_s(\mathrm{JudgeRank}_t,\ \mathrm{SystemRank}_t)\,. \tag{21}$$

   We use Spearman's rank correlation because technical fairness concerns the preservation of relative ordering rather than exact score differences, and Spearman's measure captures monotonic consistency between rankings without assuming linear relationships or comparable score scales.

2. Perceived fairness: Perceived fairness captures whether competition outcomes appear

reasonable to audiences, who tend to judge fairness based on alignment with recent performance trends rather than the underlying scoring mechanism.

To achieve in quantifying this, we first proxy recent performance by the moving average of judge rankings over the past W weeks:

$$F1_t = \rho_s(\text{JudgeRank}_t,\ \text{SystemRank}_t)\,. \tag{22}$$

and define perceived fairness as the Spearman rank correlation between recent performance and system outcomes:

$$S1_t = \rho_s(\text{RecentRank}_t,\ \text{SystemRank}_t)\,. \tag{23}$$

where $W$ denotes the window length. Higher values of $S1_t$ indicate that outcomes align more closely with recent performance trends, thereby enhancing perceived fairness.

3. Entertainment value: This refers to the extent to which a scoring system preserves outcome uncertainty and competitive intensity, thereby sustaining suspense and spectator engagement, consistent with findings in tournament and sports competition literature.

   We then operationalize entertainment value through two complementary measures that capture whether fan voting can meaningfully contribute to competitiveness and whether late-stage eliminations retain sufficient uncertainty.

   To quantify competitiveness ($E1$), we introduce small, plausible perturbations to the observed fan vote shares and recompute the elimination outcome repeatedly:

$$E1 = \frac{\#\{\text{elimination changes}\}}{R}. \tag{24}$$

   where R is the number of perturbation simulations.

   Higher values of $E1$ indicate greater responsiveness of eliminations to fan voting, while extremely low values suggest inconsequential fan influence, and values near one indicate excessive outcome fragility. Accordingly, moderate levels of $E1$ reflect a desirable balance between audience engagement and outcome robustness.

   Furthermore, since late-stage uncertainty ($E2$) is the measure of how tight elimination outcomes are in the final stages of the competition. To reflect whether results remain suspenseful rather than predetermined. We define the elimination margin in week t as $\text{Margin}_t = \text{Score}_{(2),t} - \text{Score}_{(1),t}$.

   Where (1) and (2) denote the lowest and second-lowest system scores, respectively. This

uncertainty is then defined as a normalized inverse of the average late-stage margin:

$$E2 = 1 - \frac{1}{L} \sum_{t \in \text{late}} \frac{\text{Margin}_t}{\max_s \text{Margin}_s + \eta}. \tag{25}$$

Where $L$ is the number of late-stage weeks, and $\eta$ is a small constant $(= 10^{-6})$ to ensure numerical stability. Higher values of $E2$ correspond to greater late-stage excitement, while remaining comparable and stable across seasons.

## 6.2 Proposed Scoring Rule

### 6.2.1 Normalization and Direction Alignment

Judge rankings and fan vote percentages convey performance information in opposite directions. We therefore normalize both measures onto a unified [0,1]scale, where larger values indicate better performance:

$$J_{i,t} = \text{JudgePercent}_{i,t}, \tag{26}$$

$$V_{i,t} = \text{FanPercent}_{i,t}. \tag{27}$$

This ensures that expert evaluations and audience preferences are comparable signals of contestant quality.

### 6.2.2 Adaptive Fan Weight

Instead of fixing the relative importance of judges and fans, we allow the fan vote weight to adapt based on the informativeness of the voting outcome. We measure informativeness using the dispersion of fan votes:

$$w_{\text{fan},t}^{\text{raw}} = 1 - \frac{\text{Var}(V_{\cdot,t})}{\max_s \text{Var}(V_{\cdot,s})}. \tag{28}$$

A more evenly distributed vote is interpreted as collective judgment and receives higher weight, while highly concentrated voting—potentially driven by extreme fan mobilization—is down-weighted. To ensure stability, the final fan weight is bounded:

$$w_{\text{fan},t} = w_{\min} + (w_{\max} - w_{\min})\ w_{\text{fan},t}^{\text{raw}}. \tag{29}$$

### 6.2.3 Final System Score

The system score for contestant $i$ in week $t$ is defined as

$$\text{SystemScore}_{i,t} = w_{\text{fan},t}\, V_{i,t} + (1 - w_{\text{fan},t})\, J_{i,t}. \tag{30}$$

This scoring rule directly operationalizes our design objectives: preserving technical fairness through expert evaluations, maintaining perceived fairness via adaptive weighting, and sustaining entertainment value by preventing premature outcome lock-in.

#### 6.2.4 Design–Evaluation Consistency

The proposed scoring rule and the evaluation metrics are intentionally constructed as two complementary components of a single mechanism-design framework. Each evaluation metric is explicitly designed to test a distinct mechanism embedded in the proposed scoring rule:

1. Judge–outcome correlation (F1) evaluates whether the lower bound on judge influence successfully preserves the relative ordering implied by professional evaluations, thereby validating technical fairness.

2. Recent performance consistency (S1) tests whether the week-to-week aggregation and adaptive weighting prevent abrupt rank reversals that contradict recent performance trends, which is central to perceived fairness from the audience's perspective.

3. Vote impact sensitivity (E1) directly examines the effectiveness of the adaptive fan-weight mechanism by measuring whether variations in fan votes can meaningfully alter elimination outcomes when fan voting is informative.

4. Finally, late-stage uncertainty (E2) assesses whether the imposed weight bounds and score aggregation structure prevent early outcome lock-in, ensuring that eliminations in later stages remain competitive and suspenseful.

Together, these metrics form a one-to-one validation framework that maps each core design choice of the scoring system to an observable performance criterion.

### 6.3 Results and Interpretation

To evaluate the performance of the proposed scoring system, we re-ran the competition outcomes under three alternative rules: the original rank-based rule, a percentage-based rule relying solely on fan votes, and our proposed adaptive scoring system. Each method was evaluated using the four metrics introduced in Section 5, capturing technical fairness (F1), perceived fairness (S1), vote impact sensitivity (E1), and late-stage uncertainty (E2). The results are summarized in Table under: Overall, the proposed method exhibits the most balanced performance across all dimensions. Rank-based scoring performs poorly on both fairness metrics, showing strong negative correlations that indicate systematic violations of order preservation and recent performance trends. Percentage-based scoring avoids these extremes but lacks structural safeguards, making outcomes sensitive to fan mobilization.

In contrast, the proposed system achieves the highest technical fairness (F1 = 0.355) and perceived fairness (S1 = 0.347), outperforming the percentage-based rule on both dimensions.

| Method | F1 | S1 | E1 | E2 |
|---|---|---|---|---|
| Our new method | 0.355 | 0.347 | 0.733 | 0.667 |
| percentage | 0.351 | 0.272 | 0.674 | 0.685 |
| rank | -0.704 | -0.555 | 0.994 | 0.547 |

Figure 13: Enter Caption

At the same time, it maintains a moderate level of vote impact sensitivity (E1 = 0.733), ensuring that fan participation is meaningful without rendering outcomes excessively fragile. The proposed method also preserves late-stage uncertainty at a level comparable to percentage-based scoring (E2 = 0.667), while substantially outperforming rank-based scoring.

Taken together, these results indicate that the proposed method successfully balances fairness and entertainment, offering a robust alternative to existing scoring rules that avoids the extreme behaviors of rank-only and percentage-only systems.

# 7 Overall Model Evaluations

## 7.1 Advantages

1. Bayesian inference captures uncertainty explicitly. The Bayesian component infers latent fan vote distributions and quantifies uncertainty, allowing the model to distinguish informative voting patterns from noise rather than treating observed votes as deterministic inputs.

2. Adaptive scoring translates inference into a practical mechanism. Insights from the Bayesian estimation are operationalized through an adaptive weighting rule, enabling fan influence to vary with its informational content while preserving interpretability.

3. Balanced performance across multiple objectives. The combined framework performs strongly on technical fairness, perceived fairness, and entertainment metrics, avoiding the extreme behaviors observed under rank-only and percentage-only rules.

4. Modular and extensible design. The Bayesian estimation and scoring mechanism are loosely coupled, making the framework easy to extend with alternative priors, additional behavioral assumptions, or different aggregation rules in future studies.

## 7.2 Disadvantages

1. Sensitivity to prior assumptions in early stages. When data are limited, especially in early weeks, posterior estimates may be influenced by prior choices, which can affect inferred

vote distributions.

2. Limited modeling of strategic or coordinated voting. While the adaptive weighting mitigates extreme fan mobilization, the Bayesian model does not explicitly account for strategic behavior or coordinated campaigns among voters.

3. Evaluation focuses on outcomes rather than behavioral feedback. The framework evaluates scoring outcomes but does not model how audiences might adapt their voting behavior in response to the scoring rule itself.

4. Increased computational complexity. Compared to deterministic scoring rules, the Bayesian component introduces additional computational overhead, which may limit real-time or large-scale applications.

# 8 Memorandum

**To:** Producers of *Dancing with the Stars*
**From:** Team #2629912
**Subject:** Combining Judge Scores and Fan Votes: Findings and Recommendations
**Date:** February 2026

Dear Producers,

We are pleased to share the results of our analysis regarding how different vote aggregation rules influence eliminations, final rankings, and perceived fairness in *Dancing with the Stars*. Our findings highlight systematic differences between judge evaluations and fan preferences, as well as the critical role that aggregation design plays in shaping competition outcomes.

**Key Findings**

- **Judges and fans reward different qualities.** Judges' scores primarily reflect technical performance, while fan votes reflect popularity and narrative appeal.

- **Aggregation rules materially affect outcomes.** Different vote-combination rules lead to systematically different eliminations and final rankings.

- **Percentage-based aggregation amplifies fan dominance.** Vote-percentage rules allow large fan vote margins to outweigh technical weaknesses.

- **Rank-based aggregation is more stable.** Rank-based rules limit extreme fan influence and yield more consistent outcomes.

- **Controversies concentrate under judge–fan disagreement.** Disputed outcomes mostly involve contestants with strong fan support but weak judge evaluations.

**Recommendations**

- Use rank-based aggregation as the default mechanism.
- Apply a bottom-two with judges' save as a targeted safeguard.
- Consider adaptive fan vote weighting to balance fairness and engagement.

It is worth noting that, in addition to the current system, we recommend adopting the proposed adaptive scoring model, which dynamically balances judge evaluations and fan votes based on the informational quality of audience participation. The model improves both technical and perceived fairness while maintaining excitement, avoiding the instability of fan-only systems and the rigidity of rank-only rules. Its transparent, modular design allows for practical implementation and future extensions as voting behavior evolves.

Yours sincerely,
Team #2629912

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# Report on Use of AI Tools

The team used AI-assisted tools in a limited and responsible manner to support the preparation of this submission. All final modeling decisions, mathematical derivations, data analysis, interpretations, and conclusions were developed, verified, and approved by the team.

## OpenAI ChatGPT (February 2026 version)

The tool was used for:

- Brainstorming alternative modeling approaches and checking the logical consistency of modeling assumptions.
- Assisting with code debugging and clarifying syntax-related issues during model implementation.
- Improving clarity, conciseness, and academic tone of written explanations without altering technical content or results.

All AI-generated suggestions were carefully reviewed, modified when necessary, and validated by the team to ensure correctness and originality.

## Mathematical and Programming Assistance

AI tools were occasionally consulted to explain standard mathematical concepts, optimization methods, and programming structures (e.g., loops, vectorization, regression setup) to support the team's understanding.

All code used in the final model was written, tested, and verified by the team.

No AI tools were used to automatically generate final results, fabricate data, or replace independent reasoning. The team takes full responsibility for the accuracy, integrity, and originality of the submitted work.

**Team 2629912**